

JEE MAIN REVISION MASTER: CHEMICAL KINETICS 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 34 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Chemical Kinetics - Core Application	34 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Chemical Kinetics - Core Application (Very High Yield — 34 PYQs)

Core Principle: Key problem pattern in Kcet Chemical Kinetics.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The activation energy for the reaction $X \rightarrow Y$ is 150kJ mol^{-1} . The change in enthalpy for the above reaction is -135kJ mol^{-1} ." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "For a 1st order change $R \rightarrow P$, the concentration of Reactant R changes from 0.1 M to 0.025 M in 40 minutes. The rate of re..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "For a reaction having three steps, the overall rate constant is $k = \frac{k_1 k_2}{k_3}$. The values of E_{a_1} , E_{a_2} and E_{a_3} are..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The activation energy for the reaction $X \rightarrow Y$ is 150 kJ mol^{-1} . The change in enthalpy for the above reaction is -135 kJ mol^{-1} . Then the activation energy for $Y \rightarrow X$ is

- (A) 280 kJ mol^{-1}
- (B) 285 kJ mol^{-1}
- (C) 270 kJ mol^{-1}
- (D) 15 kJ mol^{-1}

Correct Answer: None

Solution Analysis: For the reaction $X \rightarrow Y$ activation energy is $E_a(\text{forward}) = 150 \text{ kJ mol}^{-1}$ and enthalpy change is $\Delta H = -135 \text{ kJ mol}^{-1}$. Since ΔH is negative, the reaction is exothermic. This means product Y has lower energy than reactant X by 135 kJ mol^{-1} . For the reverse reaction, $Y \rightarrow X$ the activation energy will be the energy needed to go from Y up to the activated complex. Using the relation: $E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$. Substitute the values: $E_a(\text{reverse}) = 150 - (-135)$
 $E_a(\text{reverse}) = 150 + 135 = 285 \text{ kJ mol}^{-1}$